

the pad size is two-three times larger than the drill size, and for a double-side PCB, an annular ring of 0.5 mm should be maintained as minimal area for soldering.

17. Check the drill size specified in the design. Mark it as furnished or non-furnished diameter (in the case of double-side or multilayer PCBs). Furnished diameter is the diameter that is required in a furnished PCB—the manufacturer has to drill a larger diameter so as to compensate for the plating and HASL thickness. Non-furnished diameter means that whatever the diameter is specified by the designer, the manufacturer will drill the same diameter. The final diameter will be reduced depending upon the thickness of deposited copper, tin or tin-lead (HASL), during PCB fabrication.
18. Keep at least 0.2mm extra drill diameter compared to actual component lead diameter since the solder too requires some space to flow so as to make a fillet up to the top layer in a plate-through hole (PTH) PCB.
19. Ensure minimum number of drill tools in the PCB, in order to reduce the machine cycle time.
20. Define the global spacing, i.e., track-to-track, track-to-pad, pad-to-pad, track-to-via, pad-to-via and via-to-via, keeping in view the manufacturer's limitations.
21. Define the minimum and maximum trace width for different nets or connections. Otherwise, it will create errors when you perform design rule check.
22. Avoid routing traces close to the edge of the PCB, especially when the PCB has to be installed in a rack by sliding in a channel.
23. Never route traces with 90° corners. Right-angle corners not only create problems during etching of PCBs but also cause impedance mismatch in a trace, which results in rounding off of the edges of a signal. Instead of 90°, 135° or even

round corners are preferred.

24. Check the trace width depending upon the current requirement so that it can carry the current without being heated.
25. When using SMDs, never place a via hole adjacent to an SMD land area. There must be at least a thin layer of the solder mask between them. This is because when reflow soldering takes place, solder paste may flow into via holes leaving insufficient or no solder for SMD components.
26. Don't place via holes under metal components. The solder can flow through vias and short to metal components.
27. When using SMDs, do not forget to place fiducials in your design. Fiducials should be placed in individual PCBs as well as separate fiducials should be placed in the case of panels. These should be placed on the three edges of the PCB.
28. Make sure that the component notation text is not placed on a land or pad, as it will not allow the solder to flow over the complete land/pad of a component during reflow soldering.
29. Do not forget to place part number, revision code, company logo, cut mark, etc in the design.
30. Before creating NC data, write notes on PCB and copper thickness requirement, tolerances, etc.
31. When design rule check (DRC) is performed, there should not be any error or warning. It is a good practice to verify the component pins that are not connected/used in the circuit diagram.
32. Make sure that the component-to-component spacing satisfies the requirements of test and automation equipment.
33. Any labels required on the board itself should be at standard locations, especially if they are machine-readable like barcodes.
34. The panels must be properly planned and placed to ensure

maximum material utilisation for the active PCBs. All of the material will be used as one component of the PCB cost. Some of the standard panel sizes are 30.5×45.7 cm² (12×18 inch²), 40.6×45.7 cm² (16×18 inch²) and 45.7×61 cm² (18×24 inch²).

35. PCBs are often required to be stepped and repeated onto an array for assembly. Correct assembly array data should be provided to an assembler.
36. Scoring and routing are two of the most common methods used for excising printed boards or assembly panels. The easiest way to separate the board is to score each side with a diamond grinder and break the board or panel apart. The board must be square or rectangular in shape. Consult your PCB manufacturer for edge-to-edge distance before creating the panel. In addition to the technical information of the circuit and its components, the housing (fixing) mechanism of the PCB as well as the environmental operating condition should be informed to the designer. This will help the designer in improving the reliability of the assembly.

Overcoming the implementation issues

The most obvious implementation issue is to make sure that everyone uses the design-for-manufacturability (DFM) manual religiously. Continual management support is important. Some organisational or methodology changes may be necessary to optimise the effectiveness of DFM. Designers working alone or design teams composed purely of design-oriented people may not have enough technical or motivational input from the manufacturing point of view. To provide the technical input and motivation, many companies are adopting the concept of multidisciplinary design teams. ●

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